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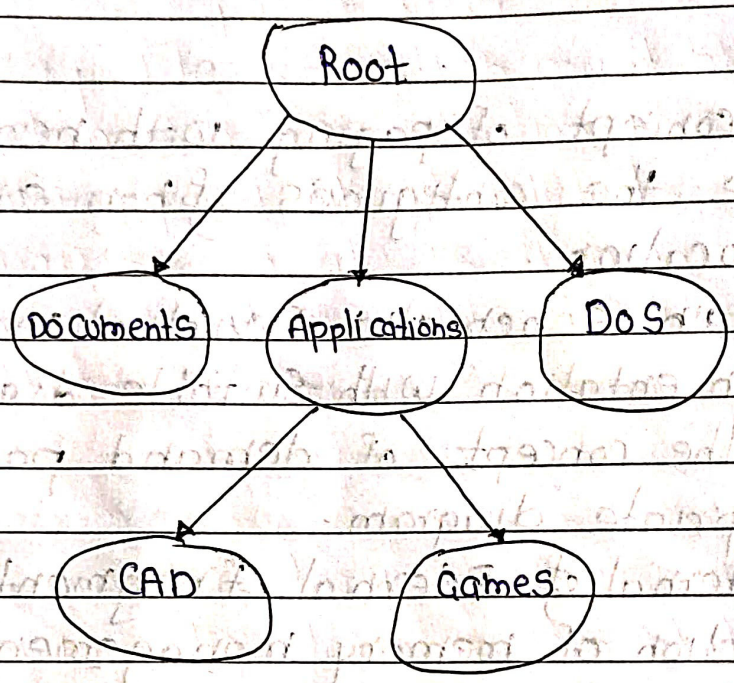
# 5. File AND DISK MANAGEMENT (UNIT-5)

Imp What is file management system? explain file system implementation in details? ( )




- The above diagram shows the structure of typical file system.
- File system is an internal part of the operating system.
- This is a traditional way of data collection & is mostly associated with secondary storage.
- File is used as the medium of giving i/p as well collecting the o/p for the most of the application.
- our computer works on stored program concept & hence program are looked up on as set of instruction grouped together in file.

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**Study material provided by: Vishwajeet Londhe**

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- The file system permits users to create data collections, called files, with desirable properties.
- Depending on the file system, a file can have its own internal structure.
- It also supports set of functions that are to be used to manage the file operations.
- Typical operations include the following.

- Create - A new file created, named & positioned within the directory structure of files.
- Delete - A file is moved out of memory file structure & is permently destroyed.
- open - Files available in the directory structure can be opened by process allowing the process to perform required functions on the files.
- closed - The file is closed with respect to a process, so that the process no longer may perform function on the file, until the process opens the files again.
- Read - A process can read all or required file data using this function.
- Write - A process can read all or required file data using this function.

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- File management system provide service to every application that user executes.
- Any application under the sun needs to co-operate with the file management system to access required file.
- Every file operation must be supported by backup & other operation like,
- Recovery : if file is deleted.
- Repairing : if file is damaged.
- owner must be able to access file by name rather numeric.

Imp

**Explain directory structure with its type discuss directory implementation ?**

**or**

**Explain the following term with respect to directory -**

- **two level directory structure with diagram**
- **tree structured directories with diagram ?**

→ There are various layers in which directories can be arranged, depending on the amount of data. It can be :

1) Single level directory structure.
2) Two level directory structure.
3) Tree structured directory.

1) **Single level Directory structure -**

This is your the simplest directory structure.

- A directory typically contains a number of

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entries, one per file.

| Data | Attributes |
|---|---|
| News | Attributes |
| Mail | Attributes |

(a)

| Data |
|---|
| News |
| Mail |

(b)

Fig. Single level directory structure.

- This type of structure has various entries; each entry contains the files name, the file attributes & the disk addresses where the data is stored or where directory holds file name & pointer to other data structure where attributes & disk addresses are found.
- when a file is opened, the OS searches its directory until it finds the name of the file to be opened.
- It then extracts the attributes & disk addresses directly from the entry & puts them in a table in main memory.

2) Two Level Directory Structure

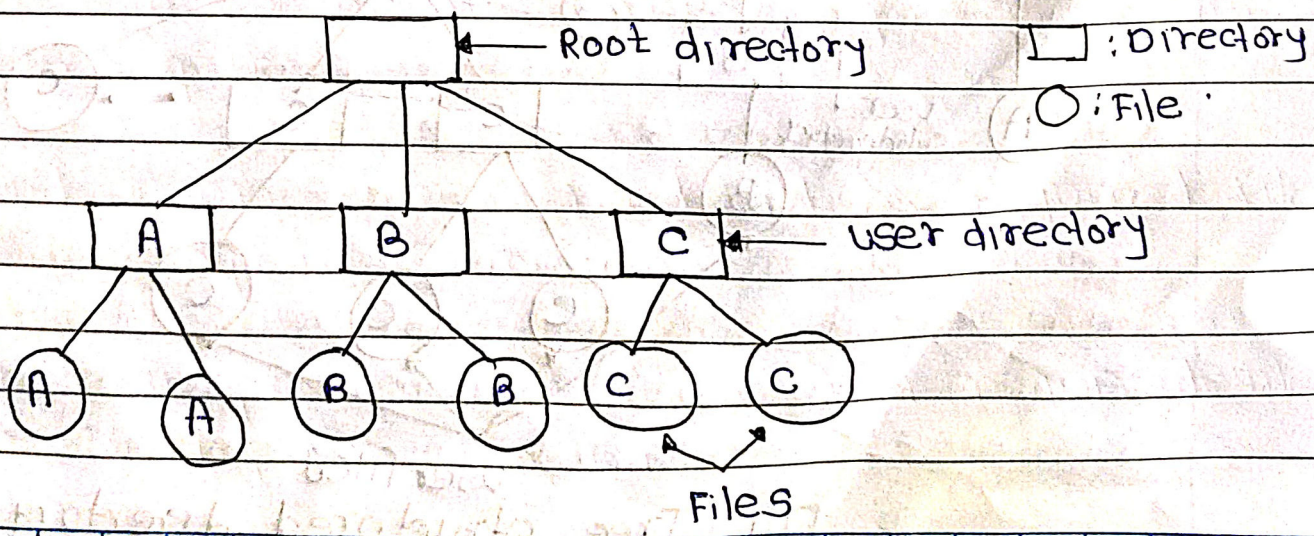


Fig - Two level directory structure.

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- In order to overcome the disadvantages of single level structure, another idea is to have a single directory for all files in the entire system is to have one directory per user.
- In this two level directory, each user has his/her own user file directory (UFD).
- When a user job starts, the systems roots directory is searched.
- The root directory is indexed by user name which points to user directory.
- A two level directory structure can be thought of as a tree or an inverted tree of height 2. The root of the tree is root directory.

3) Tree Structured Directory -

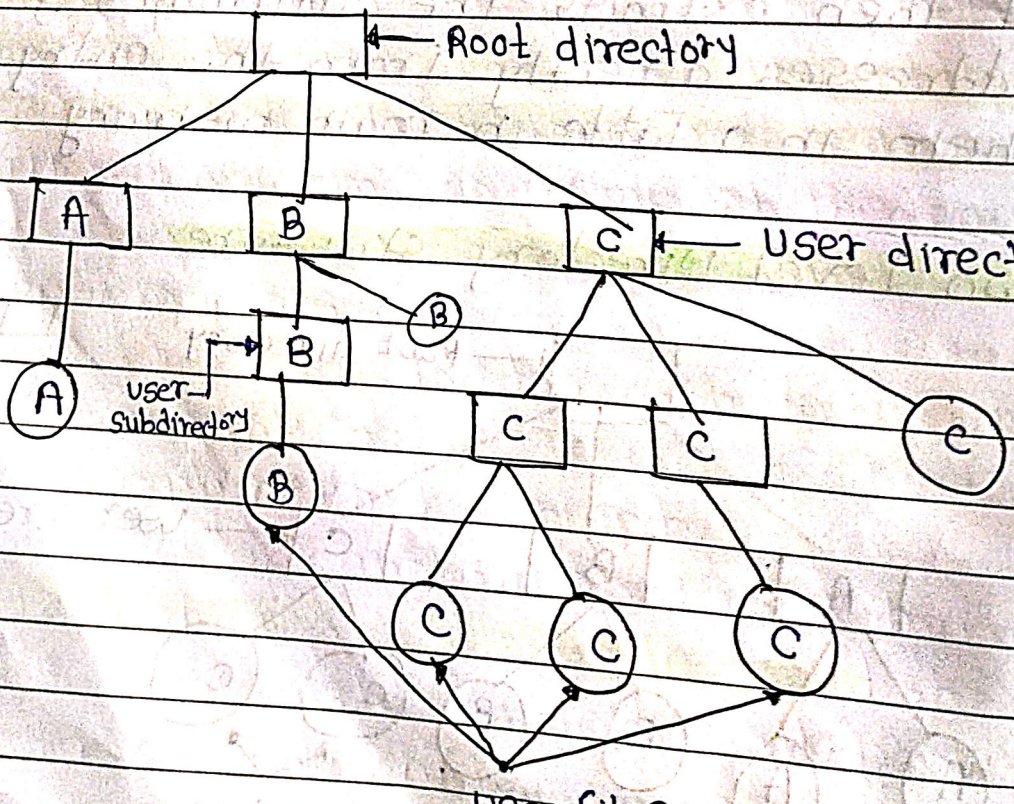


Fig. Tree structured directory

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- An instresting, policy decision in a tree structured directory is how to handle the deletion of a directory.
- If a directory is empty, its entry in its containing directory can be deleted.
- But if the directory to be dpleted is not empty, but contains several files or sub-directories.
- With a tree structured directory system, users can access, in adition to their files, the files of other user.

Imp (Pool) (Nov-14)

**Why I/O buffering is necessary? State & explain different I/O buffering techniques.**

or

**Explain I/O buffering mechanism.**

or

**Describe I/O buffering in details?**

→ A buffer is a memory area that stores data while they are transferred between two devices or between a device and an application

- There are three reasons for buffering technique to be implemented.
- Speed of mismatch between the producer & consumer of data stream.
- Another reason is to support copy semantics for application I/O.
- Let us explain all three above explained reasons with example:

• Speed Mismatch - For example, that a file is

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being received via modem for storage on the hard disk.
- The modem is above a thousand time slower than the hard disk.
- So a buffer is created in main memory to accumulate bytes that are received from modem.
- When an entire buffer of data has arrived, the buffer can be written to disk in a single operation.

• **Different Data Transfer sizes -**

Such a disparity is common in computer networking, when buffer are used widely for fragmentation & reassembly of message.
- At the sending side, a large message is fragmented into small network packets.
- The packets are sent over the networks, the receiving side places them in a reassembly buffer to form an image of the source data.

• **Copy Semantics -**

Third reason to use buffering is copy semantics.
- Let us explain this concept.
- Suppose that an application has a buffer of data that it ~~wishes~~ wishes to write to disk.
- It calls the write system call providing a pointer to the buffer & an integer

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specifying the number of bytes to write. After the system call returns, what happens if the application changes the contents of buffer: with Copy Semantics the version of the data written to disk is guaranteed to be version at the time of the application system call, independent of any subsequent changes in the application's buffer.

## DMA (Direct memory Access)

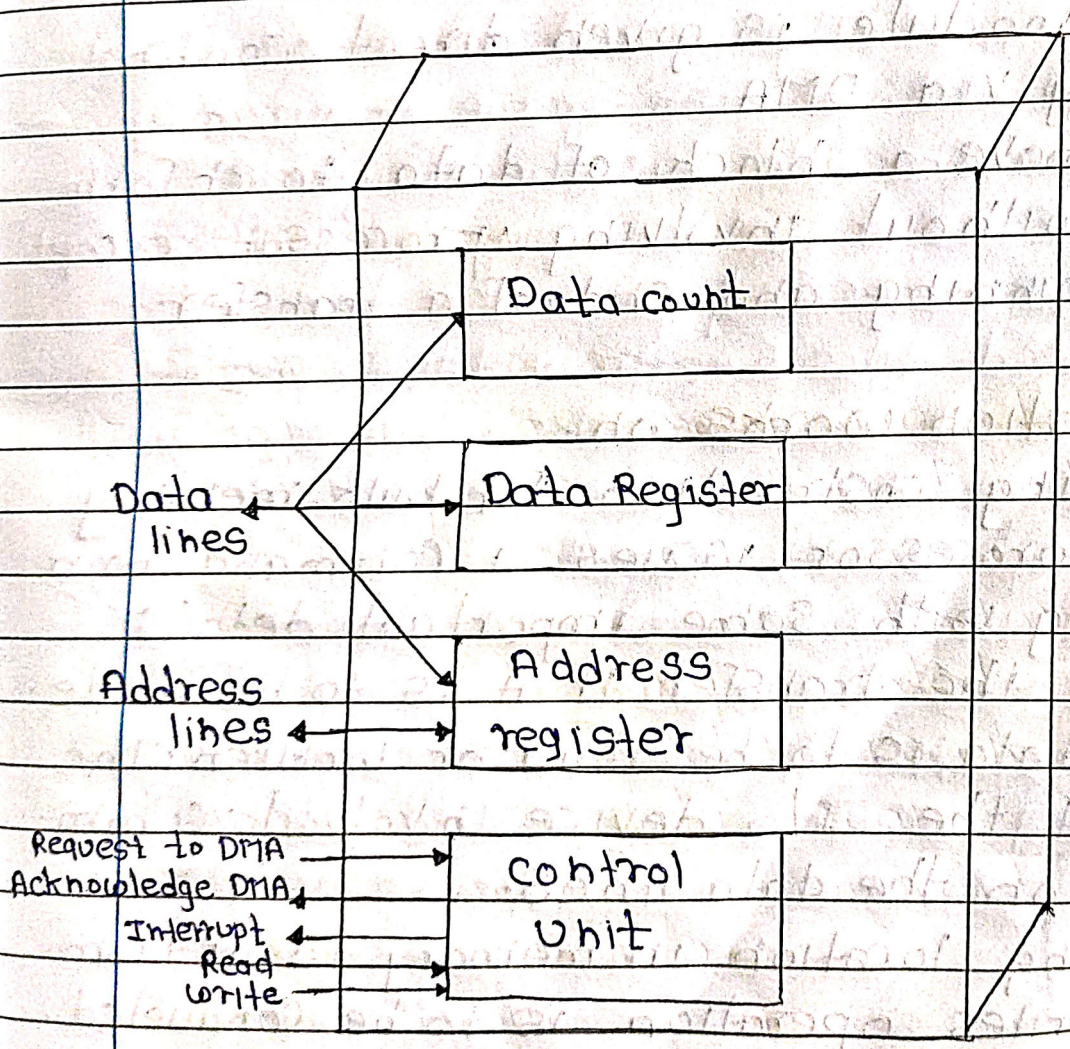


Fig: DMA block diagram

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- The I/o functions have evolved as the computer system is evolving.
- The processor directly controls a peripheral device.
- A controller is added.
- The processor uses programmed I/o without interrupt.
- Same configuration as in step b is used but interrupts are involved.
- The processor need not spend time waiting for an I/o operation to be performed.
- The I/o module is given direct control of memory via DMA
- It can move a block of data to or form memory without involving processor, except at the beginning and end of a transfer.

DMA operation works on.

- on receiving instruction for bulk memory transfer processor issues a command to a DMA along with some important details regarding the transfer that is to be made.
- whether data is to be read or written. The address of the I/o device involved, communication on the data lines.
- the sharing location in memory to or from read or writes operation is to be conducted.
- This is made available through DMA address register.

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- The number of words to be read or written this is conveyed through DMA Count register.
- processor relinquishes the bus control to DMA & continues with other work.
- DMA performs this task using different approaches like.

1) Burst mode - Bus control is given back any after completion work.
2) cycle stealing mode - Bus control is acquired in between cycles when bus is idle.
3) Bus Interleaved mode - Bus control is obtained interrupt in between.

- when the transfer is complete the DMA module sends an interrupt signal to the processor.
- The type of architecture also influences the DMA operation.

A) Single Bus, Detached DMA Controllers:

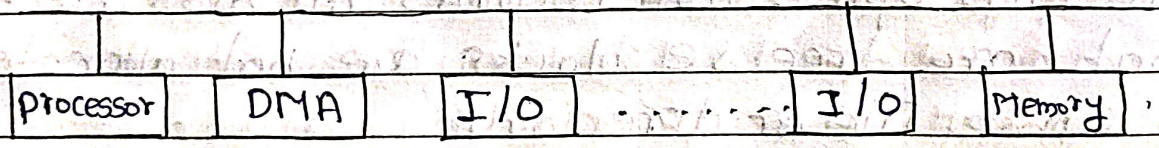


Fig. Single bus, detached DMA controller

Each Transfer uses Bus Twice

- DMA to memory
- CPU is suspended once.
- I/O to DMA then DMA to memory.
- CPU is suspended Twice.
- Controller may support > 1 device

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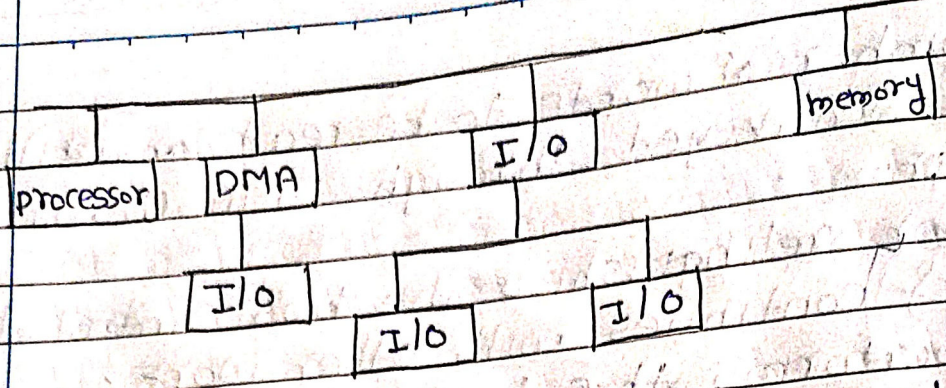


single-bus, integrated DMA-I/O

Imp ★ Explain free space management technique
or
Free space management (FSM)

→ Files are normally stored on disk.
- So management of disk space is a major concern for the file system designers.
- Since there is a limited amount of disk space, it is necessary to reuse the space from deleted files for new files, if possible
- To keep track of free disk space, the system maintains a free space list.
- The free space list records all disk blocks that are free. se blocks are not allocated to same file or directory.
- For this stuff, let get back to the free list concept, when a file is deleted its disk space is added to the free space list.
- The Free list can be implement in many ways.

1) Bit vector - Usually free space list is implemented as a bit vector.

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- Each block is represented by 1 bit.
- If the block is free, the bit is 1. If the block is allocated, the bit is 0.
- Bit vector are inefficient unless the entire vector is kept in main memory
- Keeping it in memory is possible for smaller disk, but not for larger ones.

- Linked List-

Another approach is to link together all the free disk blocks, keeping a pointer to the first free block in a special location

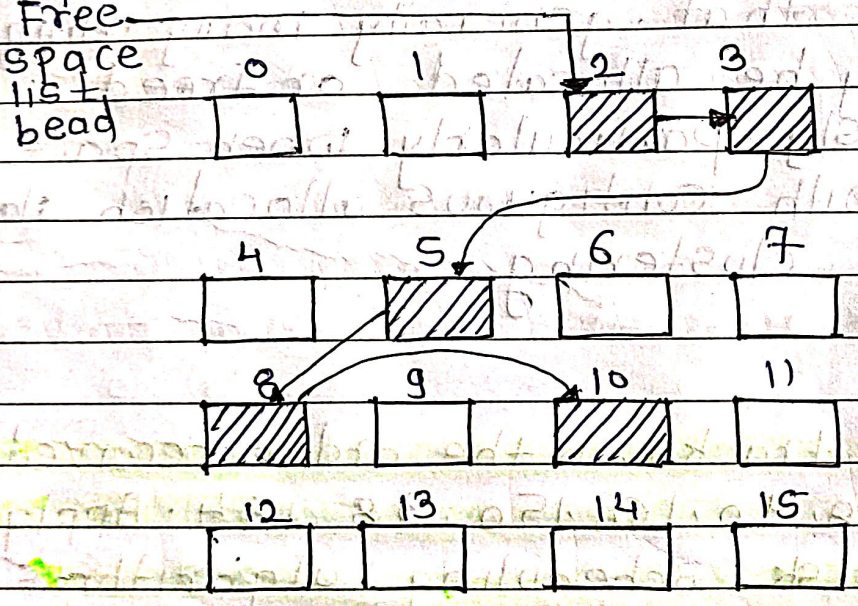


Fig. Link List

- on the disk & caching in it memory
- First block contains a pointer to the next free disk block & so on.
- The os simply needs a free block so that it can allocate that block to a file, so the first block in the free list is used

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- Grouping -
  Another approach is to store addresses of n free blocks in the first free block.
  - The first n-1 of these block are actually free.
  - The last block contains the addresses of another n free block & so on.
  - Advantage of this approach is that addresses of larger number of free blocks can be found quickly.

- Counting -
  Last approach is to take advantage of the fact that generally many contiguous blocks may be allocated or freed simultaneously, particularly when space is allocated with contiguous allocation algorithm or through clustering.

Imp The required tracks, in the order received are: 55, 58, 39, 18, 90, 160, 150, 38, 184. Apply the following disk scheduling algorithms starting track at 100.

1) FCFS
2) SSTF
3) SCAN
4) C-SCAN

→ 1) FCFS -

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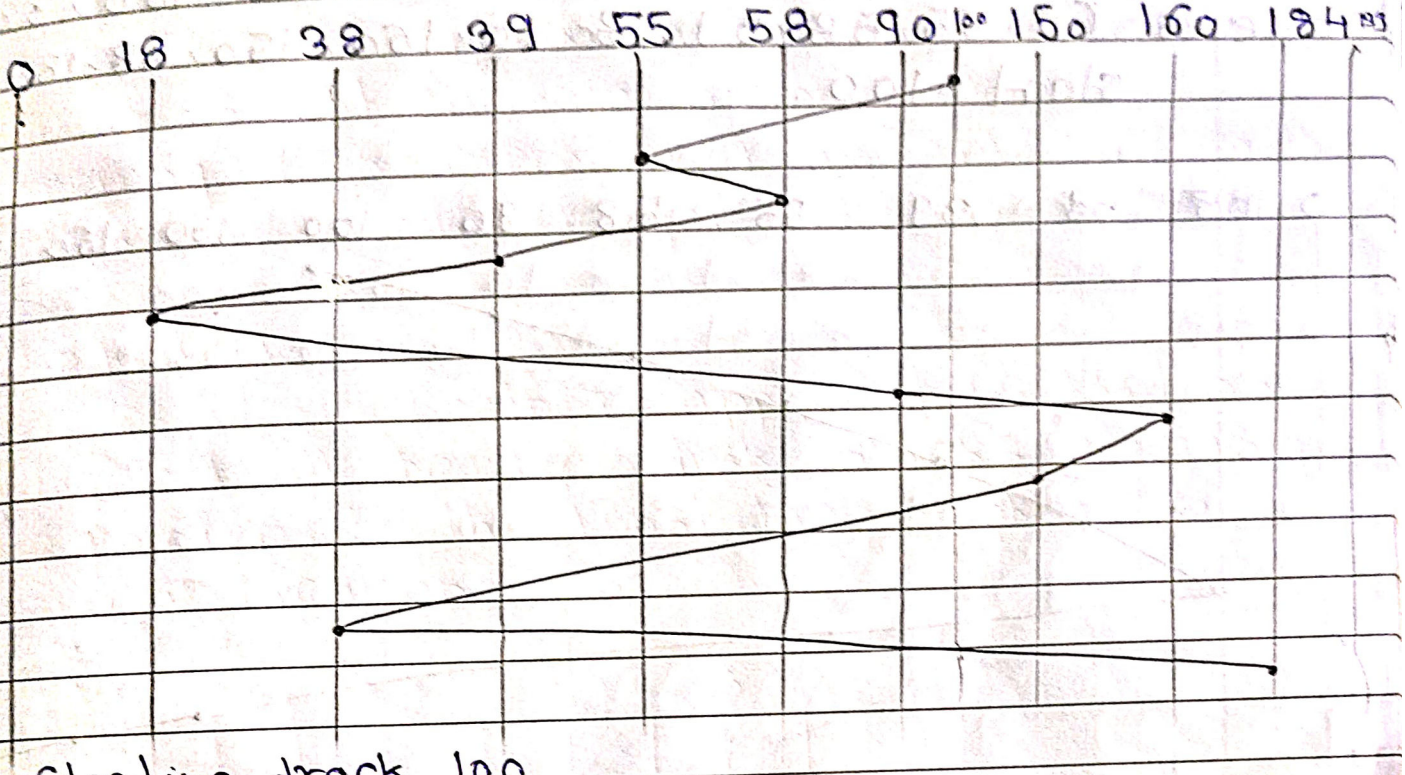
Request-(55, 58, 39, 18, 90, 160, 150, 38, 184).
queue-(0, 55, 58, 39, 18, 90, 160, 150, 38, 184, 199)
start track -100




Starting track 100

| Next Track Accessed | Number of Track Traversed |
|---|---|
| 55 | (100 - 55) = 45 |
| 58 | (58 - 55) = 3 |
| 39 | (58 - 39) = 19 |
| 18 | (39 - 18) = 21 |
| 90 | (90 - 18) = 72 |
| 160 | (160 - 90) = 70 |
| 150 | (160 - 150) = 10 |
| 38 | (150 - 38) = 112 |
| 184 | (184 - 38) = 146 |

Average seek length = $\frac{45+3+19+21+72+70+10+112+146}{9}$

$= \frac{498}{9} = 55.3$

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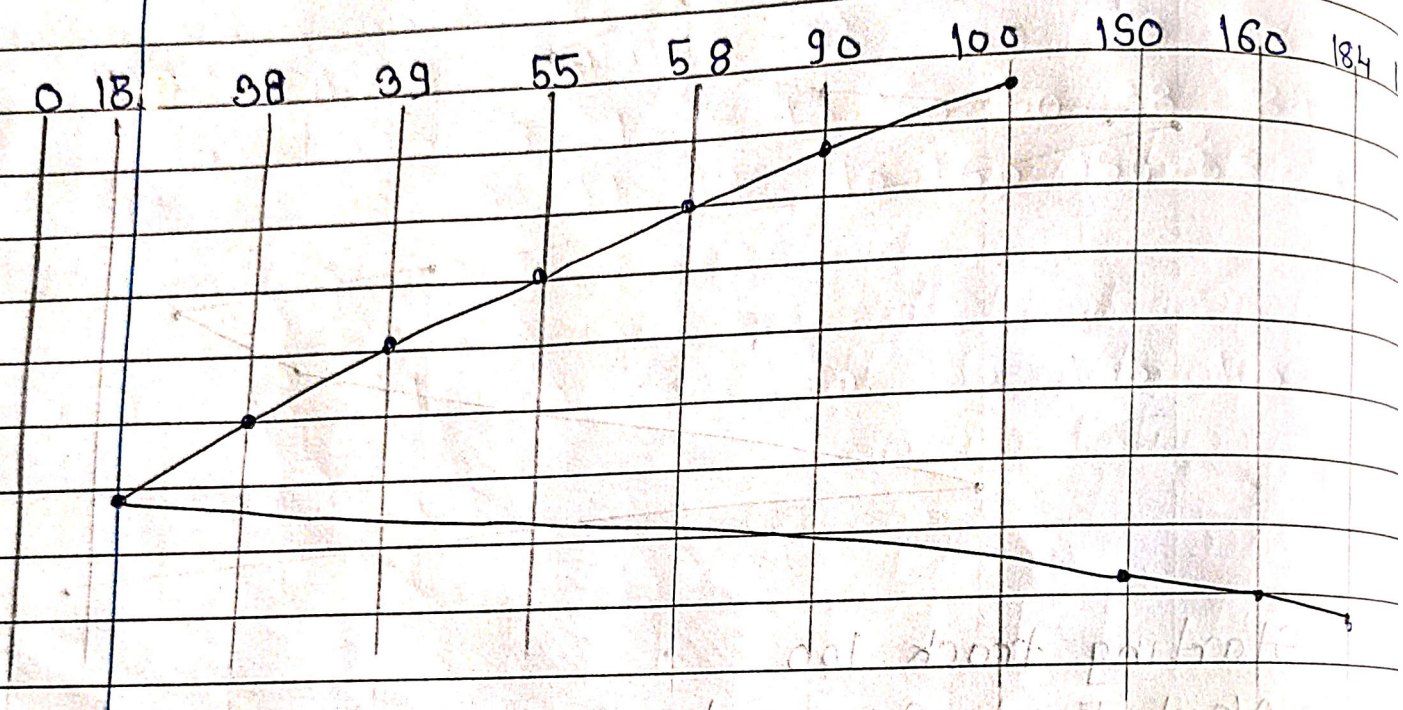
SSTF - Shortest Seek Time First scheduling

Request - (55, 58, 39, 18, 90, 160, 150, 38, 184)

queues - (0, 55, 58, 39, 18, 90, 160, 150, 38, 184, 199)

start - 100




2) SSTF

| Next Track accessed | Number of Track Tr |
|---|---|
| 90 | (100-90) = 10. |
| 58 | (90-58) = 32 |
| 55 | (58-55) = 3. |
| 39 | (55-39) = 16 |
| 38 | (39-38) = 1 |
| 18 | (38+18) = 20 |
| 150 | (150-18) = 132 |
| 160 | (160-150) = 10 |
| 184 | (184-160) = 24 |

Average Seek length = $\frac{10+32+3+16+1+20+132+10+24}{9}$

=

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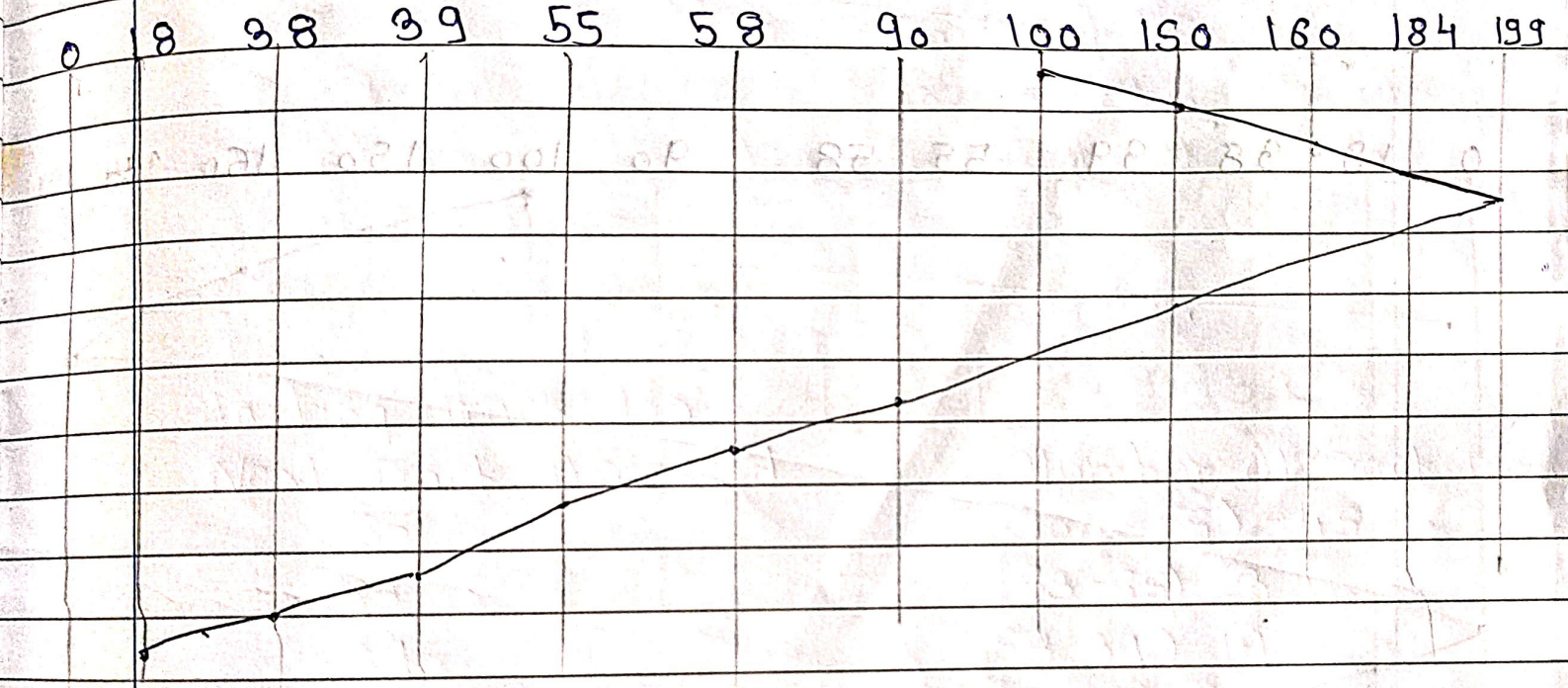
$= \frac{247}{9}$

$= 27.5$

3) SCAN -

Request - (55, 58, 39, 18, 90, 160, 150, 38, 184)

queues - (0, 55, 58, 39, 18, 90, 160, 150, 38, 184)




SCAN

| Next Track access | No of Track Traversed |
|---|---|
| 150 | (150 - 100) = 50 |
| 160 | (160 - 150) = 10 |
| 184 | (184 - 160) = 24 |
| 90 | (184 - 90) = 94 |
| 58 | (90 - 58) = 32 |
| 55 | (58 - 55) = 3 |
| 39 | (55 - 39) = 16 |
| 38 | (39 - 38) = 1 |
| 18 | (38 - 18) = 20 |

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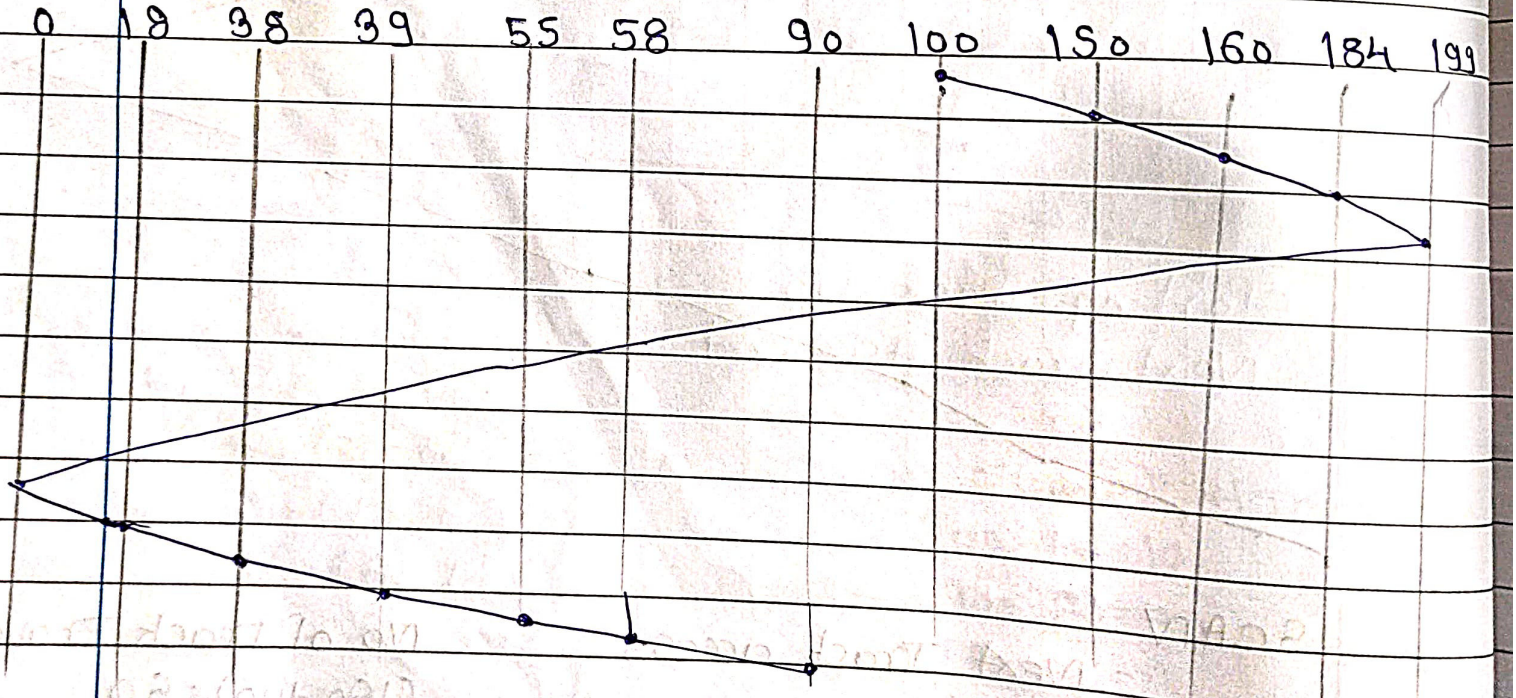
Average seek length = $\frac{50+10+24+94+32+03+16+01+20}{9}$

$= 27.8$

4) C-SCAN - (circular-scan)

Request - (55, 58, 39, 18, 90, 160, 150, 38, 184)

queue - (0, 55, 58, 39, 18, 90, 160, 150, 38, 184, 199)



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TOPIC | Page No.
Date / /

C-SCAN

| Next Track Accessed | Number of Track Traversed |
|---|---|
| 150 | (150-100) = 50 |
| 160 | (160-150) = 10 |
| 184 | (184-160) = 24 |
| 18 | (184-18) = 166 |
| 38 | (38-18) = 20 |
| 39 | (39-38) = 1 |
| 55 | (55-39) = 16 |
| 58 | (58-55) = 3 |
| 90 | (90-58) = 32 |

Average Seek length =

$$\frac{50+10+24+166+20+1+16+3+32}{9}$$

$$= 35.8$$